# *FTM-Sense*: Robust Sensor-free Occupancy Sensing Leveraging WiFi Fine Time Measurement

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## ABSTRACT

Occupancy detection is vital in optimizing smart building applications, such as automatic heating/cooling, lighting systems, and energy management. Conventional occupancy sensing approaches have limitations, such as inadequate performance with stationary occupants, susceptibility to environmental factors, and limited adaptability to various indoor environments. To address these limitations, we propose *FTM-Sense*, a real-time adaptable sensor-free occupancy detection system by employing the WiFi fine time measurement (FTM) protocol. Our proposed system leverages the variations in FTM measurements caused by human body presence for occupancy detection purposes. *FTM-Sense* has the advantage of accurately detecting static occupants, and demonstrating robustness and adaptability to various indoor environments. Experimental results show that *FTM-Sense* achieves a 97.72% overall accuracy, with 97.68% accuracy in detecting static occupants. Furthermore, the system dynamically adapts to room configuration changes within 80 seconds without compromising detection performance. *FTM-Sense* is tested in both residential and office buildings, collecting over 17 hours of data in seven different rooms with varying sizes and characteristics while also evaluating three different materials for blockage in non-line-of-sight (NLOS) scenarios. This research presents a promising solution for reliable and adaptable sensor-free occupancy detection, contributing to energy management and building cost reduction.

## CCS CONCEPTS

• **Computer systems organization → Embedded systems, Sensor networks**.

## KEYWORDS

WiFi, Fine Time Measurement (FTM), occupancy detection, sensor-free, wireless sensing, human detection

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## 1 INTRODUCTION

Precise indoor occupancy detection is a challenging problem, yet a reliable and robust solution can lead to many applications, such as automatic lighting, heating/cooling control, and efficient evacuation planning in smart buildings [5, 6, 17, 23, 24]. Various occupancy detection approaches have been proposed to address this challenge [7, 18, 25, 26, 30, 31]. Traditional methods like passive infrared (PIR) sensors and cameras have limitations, including inadequate performance with static occupants and privacy concerns. Indoor air quality (IAQ) sensors, while preserving privacy, suffer from slow detection times and confusion caused by ventilation, compromising their reliability for detecting occupants.

With the rise of low-cost Internet of Things (IoT) devices and advancements in low-power radio frequency (RF) technology, new opportunities have emerged for occupancy detection. Recent studies explored RF-based approaches to address the limitations of the above methods [9, 11, 12, 32] in occupancy sensing. Among RF-based approaches, researchers identified WiFi-based approaches promising due to the widespread availability of WiFi in buildings. However, conventional WiFi-based methods, such as those using received signal strength (RSS) and channel state information (CSI), lack adaptability to changes in the physical environment and are susceptible to performance degradation due to interference from other WiFi signals [11, 12, 19, 36]. While these approaches are considered sensor-free, deploying them in a new environment necessitates additional complex processing efforts to ensure their suitability in the new surroundings.

Wang, Nikseresht et al. [32] introduced a sensor-free occupancy detection approach utilizing a WiFi protocol called fine time measurement (FTM) introduced in IEEE 802.11-2016 [2]. WiFine demonstrates the feasibility of using FTM round-trip time (RTT) measurements between WiFi devices for indoor occupancy detection, showcasing higher accuracy and lower data rates compared to a similar CSI-based approach. However, they only tested their system in two simple scenarios involving occupants walking and sitting in a conference room, covering the most straightforward cases to demonstrate the concept.

The core concept behind employing FTM for occupancy detection is to leverage the fluctuations in RTT measurements between two WiFi-enabled devices caused by multipath reflections from objects, including the human body, within an indoor space. Therefore, we develop *FTM-Sense* by leveraging the power of FTM to tackle constraints in current occupancy sensing methods, with a specific focus on detecting static occupants and ensuring seamless adaptability to diverse indoor spaces. These spaces encompass a wide range of possibilities, ranging from entirely new environments to the same room with varying configurations. For that purpose, we developed four baseline models based on 3 hours of data we collected in a single meeting room and assessed *FTM-Sense* in extreme scenarios that usually happen in indoor spaces without retraining those baseline models.

*FTM-Sense* unlocks the potential of utilizing existing WiFi infrastructure with off-the-shelf single-antenna PCB devices, making it highly accessible and easily deployable. Its exceptional performance in diverse real-world scenarios, without the need for training, makes it a versatile solution applicable throughout the building. By eliminating the necessity of deploying various sensors tailored to different spaces, *FTM-Sense* offers a cost-effective and efficient occupancy detection solution for smart buildings.

To evaluate *FTM-Sense* performance, we collected extensive experimental data in diverse indoor environments, including residential and office buildings, totaling over 17 hours of data across seven different rooms with distinct characteristics. Additionally, we tested the system in non-line-of-sight (NLOS) scenarios using various blockage materials to assess the system's robustness. Experimental results demonstrate *FTM-Sense* as a promising alternative for indoor occupancy detection, achieving an accuracy of 97.72% overall without considering the scenario, and 97.68% accuracy, specifically in detecting static occupants. Moreover, the system adapts to room configuration changes within 80 seconds without compromising occupancy detection performance. Its consistent and accurate performance in diverse indoor environments, including NLOS scenarios, reinforces its potential for widespread adoption in smart buildings without needing external sensor deployment. *FTM-Sense* offers several desirable characteristics as an occupancy detection system, including:

- **Adaptability to new environment:** *FTM-Sense* adapts to new indoor environments, eliminating the need for model training or extra effort in each environment. This capability ensures the system can be easily deployed in various spaces, which makes it a practical solution for buildings.
- **Static occupant detection:** The system can detect static occupants, an attribute that is essential in scenarios where occupants may remain stationary for extended periods, such as offices or meeting rooms.
- **Detecting through wall:** Our system can accurately detect the presence of occupants even through walls, enhancing its usability and extending its applications to scenarios where direct line-of-sight is impossible.
- **Energy and cost-efficient:** The system is designed to be cost-effective and energy-efficient, leveraging very low-cost FTM-enabled devices (ESP32s2). This approach ensures the system remains accessible and feasible for widespread deployment in smart buildings.
- **Privacy-preserving:** The sensor-free nature of *FTM-Sense* ensures that privacy concerns among occupants are addressed effectively.

By leveraging these characteristics, *FTM-Sense* presents a viable solution to the challenges faced by current occupancy sensing techniques, unlocking new possibilities for efficient and reliable occupancy detection in various indoor environments. The data supporting this study's findings are available on request to the authors.

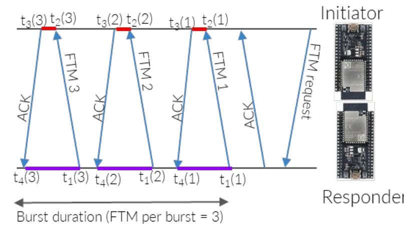



**Figure 1: Overview of WiFi fine time measurement (FTM), a two-way time-of-flight based ranging protocol that provides a higher resolution time measurement from the PHY layer.**

## 2 BACKGROUND

Fine Time Measurement (FTM) is a WiFi protocol standardized in IEEE 802.11mc amendment [2]. It facilitates distance measurement between two WiFi devices by exchanging packets and computing the round-trip time (RTT) of the packet's flight. FTM enables high-resolution time measurements, providing a valuable tool for ranging and localization applications in WiFi-based systems. Figure 1 provides an overview of the FTM protocol, where a WiFi device referred to as the *initiator* initiates the process by scanning for nearby FTM-enabled devices. When the initiator detects another FTM-enabled device, it sends an *FTM request*, including an *acknowledgment (ACK)* request to the detected device. The detected device decides to accept or decline participation in the FTM process. If it accepts the request, it becomes the *responder* and engages in a series of packet exchanges (FTM, ACK) with the initiator, forming a *burst*. The initiator specifies the number of messages in a burst.

The RTT computation is based on timestamps recorded during the burst packet transmission and reception. Assuming the burst contains $n$ FTMs, the RTT is calculated as:

$$RTT = \frac{1}{n}\sum_{k=1}^{n}((t_4(k) - t_1(k)) - (t_3(k) - t_2(k))) \tag{1}$$

The protocol accounts for the initiator processing time, highlighted in purple, by subtracting $(t_3 - t_2)$ from the total round-trip time, $(t_4 - t_1)$, highlighted in red in Figure 1. This precise RTT measurement allows for accurate distance calculation between WiFi devices and enables various applications, including occupancy sensing in indoor environments. The distance between the two devices is calculated by multiplying the speed of light with half of the RTT, denoted as:

$$d = \frac{1}{2} \times RTT \times c \tag{2}$$

The IEEE standardization of FTM and its integration into the firmware of numerous WiFi chips present substantial opportunities for wireless sensing in indoor spaces. FTM's ability to detect subtle changes in the multi-path reflections within the environment makes it a powerful tool for accurate ranging and localization applications. Leveraging FTM in occupancy sensing exploits these advantages, offering a low-cost and low-power solution for indoor occupancy detection.

## 3 RELATED WORKS

### 3.1 Non RF-based Techniques

Current non-RF-based approaches for occupancy sensing have shown some limitations that hinder their widespread adoption in certain scenarios. For instance, vision-based systems [28, 29] offer high-resolution occupancy detection but raise privacy concerns due to their reliance on capturing visual information, making them less desirable for privacy-sensitive environments [3]. Additionally, these systems require higher computational resources for image processing, making them less suitable for resource-constrained environments. On the other hand, Passive infrared (PIR) sensors [15, 33], are widely used in current smart buildings due to their low power consumption and cost-effectiveness. However, they cannot detect motionless occupants, their coverage is restricted, and at least one sensor is required per room as they cannot see through walls [33]. Utilizing indoor air quality (IAQ) sensors [7, 27], is another non-RF-based approach, detecting occupants based on air quality changes such as CO2, temperature, pressure, and humidity. While this approach does not require additional sensor deployment, it exhibits slow response times to occupancy changes and can be affected by other environmental factors, potentially leading to inaccurate results.

Given these limitations, there is a growing need to explore RF-based approaches for occupancy sensing. RF-based systems offer non-intrusive and privacy-preserving solutions by analyzing the impact of wireless signals on propagated signals without requiring any physical attachment to individuals. These systems can provide real-time occupancy detection and are less affected by environmental factors. The potential of RF-based approaches allows researchers to aim to address the limitations of current non-RF-based techniques and unlock new possibilities for efficient and reliable occupancy sensing in various indoor environments. Table 1 provides an overview of the advantages and limitations of different sensing technologies, highlighting the potential benefits of RF-based methods for occupancy detection.

**Table 1: Comparison of Occupancy Sensing Techniques**

| Technologies | Advantages | Limitations |
|---|---|---|
| Vision-based [3, 28] | High-accuracy | Privacy invasive; Intensive computation; Expensive |
| PIR [15, 33] | Low-cost; Low-power; Easy installation | Not accurate in static scenarios; Requires additional sensors deployment |
| IAQ [10, 27] | Privacy-preserving; No need for additional sensor deployment | Delayed response to occupancy changes; Affected by other environmental factors |
| RF-based [11, 21, 32, 36] | Non-intrusive; Privacy-preserving; Real-time detection | Affected by environmental factors; Requires careful RF design |

**Table 2: RF-based Occupancy Sensing Techniques**

| Approach | Advantages | Limitations |
|---|---|---|
| BLE [9, 14, 34] | Utilizes ubiquitous BLE-enabled devices; Low-power; Low-cost | Requires occupants' cooperation or additional sensors for adaptability. |
| WiFi-RSS [35, 35] | Uses existing WiFi infrastructure | Requires occupants' cooperation for adaptability. |
| WiFi-CSI [11, 21, 36] | High accuracy; No user cooperation needed; Uses existing WiFi infrastructure | Limited adaptability to new environments; High data rate; Interference with other WiFi signals |
| *FTM-Sense* | High accuracy; No user cooperation needed; Uses existing WiFi infrastructure; Low data rate; Low-power; Low-cost | The system may get confused by occupants passing near the room wall from the outside. |

## 3.2 RF-based Techniques

The evolution of low-power RF signals allows employing them for various human sensing applications ranging from vital sign detection to collision prediction in contact sports [4, 22], and occupancy sensing is one of the most popular use cases. Researchers have explored using BLE for occupancy detection in recent years due to its ubiquity in most commercial mobile devices [14]. Occupant positions can be estimated using the received signal strength indicators (RSSI) from multiple BLE-enabled devices in indoor spaces [34]. BLECS proposed a system that utilizes existing BLE-enabled devices in the room without requiring occupant cooperation [9]. However, this system struggles to accurately detect occupants when room configurations change without assistance from other sensors.

WiFi has emerged as a highly promising RF-based alternative for occupancy sensing, given its prevalence in smart buildings and extensive research exploring its feasibility for occupancy detection. For instance, WinIPS presented an occupancy scheme utilizing RSS from existing WiFi infrastructure and mobile devices [35], but it relies on occupants carrying a device for effective detection, limiting its practicality in some scenarios. Depatla et al. proposed a device-free system based on RSS measurements [12], but it lacks the adaptability to new environments. On the other hand, channel state information (CSI)-based methods attempt to sense occupants' presence without user cooperation [21, 36]. These approaches measure the impact of occupants' presence and movement on the WiFi propagation channel. However, most CSI-based approaches use conventional WiFi routers and older CSI platforms, requiring specific WiFi modules such as Intel 5300 NIC [16] or Qualcomm Atheros WiFi chip [8]. While Wi-Cal [11] demonstrated the potential of low-cost single-antenna devices like the ESP32S2 chip [13], it still faces challenges in adapting to new environments, necessitating dynamic noise reduction algorithms.

WiFine [32] recently demonstrated the viability of using fine time measurement (FTM), a newer WiFi protocol, for occupancy sensing (FTM). WiFine illustrated the great potential of FTM for indoor sensing as it requires less complex pre-processing compared to CSI and enables detection with lower data rates. Notably, we leverage FTM for occupancy detection in this study, ensuring the system's robustness and adaptability to new indoor environments without compromising accuracy. By utilizing FTM's advantages, *FTM-Sense* presents an innovative and effective solution for indoor occupancy sensing in various smart building applications without the need for extra training processes in different challenging occupancy cases like static occupant presence. Table 2 presents the benefits and boundaries of various RF-based occupancy sensing technologies.

# 4 SYSTEM DESIGN

## 4.1 Overview of *FTM-Sense*

*FTM-Sense* deployment has five key components, including round-trip-time (RTT) measurement, fine time measurement (FTM) data extraction, FTM data bundling, feature extraction, developing models, and occupancy detection. Figure 2 presents a comprehensive overview of our occupancy sensing system, and Figure 3 illustrates the system deployment in a small meeting room at a smart office building, with two ESP32S2 devices, one functioning as the initiator and the other as the responder.

To minimize interference with active wireless traffic and ensure robust data collection, we gather FTM data over ten channels within the 2.4 GHz ISM band, rather than relying on a single channel [32]. This approach guarantees robust data collection while maintaining compatibility with the building's wireless infrastructure. Subsequently, the system pre-processes the collected FTM data, combining measurements from multiple channels into an observation. Each observation comprises ten variables, with each variable representing the measured RTT value from the same channel across different observations. This aggregation enhances the accuracy and reliability of occupancy detection.

After pre-processing, we extract three essential features from the time-series FTM data, following the methodology introduced in WiFine [32]. We specifically utilized these futures to capture relevant characteristics related to the signal variation caused by human presence, thus enabling the machine-learning classifiers to accurately determine the occupancy status of the monitored space.

Our system achieves robust occupancy detection by employing ESP32S2 devices, collecting data over multiple channels, and extracting informative features. This approach ensures seamless integration with the existing wireless infrastructure, leading to improved accuracy and facilitating the implementation of various smart building applications, such as automated lighting and HVAC control, based on real-time occupancy information.

## 4.2 *FTM-Sense* Occupancy Detection Module

The system initiates the occupancy detection process by collecting time series FTM measurements over ten different channels using a single initiator-responder pair. Each raw data sample contains the timestamp, responder's MAC address, channel number, average RTT value, estimated distance between the pair, and raw FTM readings. Then, a pre-processing step combines FTM measurements over a round of multiple channels into an observation. This ensures that baseline measurement across different channels does not influence the processing results.

The extracted time series FTM data is denoted as $d_i = [d_{i1}, d_{i2}, ...., d_{ic}]$, where $i$ is the observation index, and $c$ represents the total number of channels used for distance measurement. The data is bundled into fixed-size time windows. Each data bundle is represented as $D^w = [D_1^w, D_2^w, ...., D_c^w]$, where $D^w$ is the data for a particular time window $w$. The sliding windows result in overlapping data bundles for each bundle, having $w - 1$ samples overlapping with its next bundle, resulting in $i - 4$ bundles for $i$ observations.

Wang and Nikseresht et al. [32] identified three features of variance (var), standard deviation (std), and root mean square (rms) for FTM-based occupancy detection. These features have proven to be informative and valuable for the application at hand. The key rationale behind selecting these features is their significance in capturing variations in the FTM signal caused by human body motion, which plays a pivotal role in FTM-based occupancy detection. Therefore, we extract these crucial features from the data bundles. Subsequently, we feed extracted features into four classification algorithms of random forest (RForest), k-nearest neighbors (KNN), XGBoost, and logistic regression (LR) to develop our baseline models.

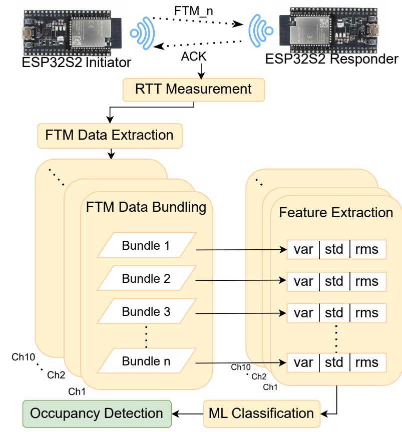


**Figure 2: System Overview**

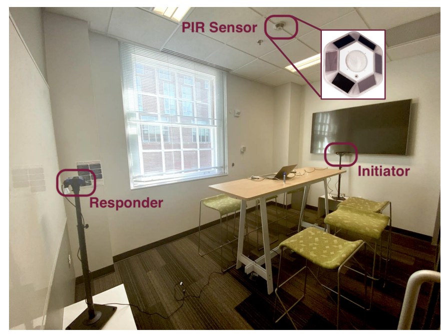


**Figure 3: Sample deployment of single antenna FTM enabled WiFi devices.**

# 5 IMPLEMENTATION

We implement a prototype of *FTM-Sense* system using two standard ESP32S2 boards. These boards support FTM measurements on the 2.4 GHz ISM band through their built-in single PCB antenna. Importantly, our implementation does not necessitate modifying the WiFi protocol firmware. Instead, all modifications are made at the application layer, ensuring easy adaptability to other platforms.

Our system comprises an initiator and a responder, which conduct the FTM process on a single channel before the initiator requests a channel switch. Subsequently, both devices switch to the next available channel, continuing the FTM process. To obtain comprehensive data, we collect FTM measurements from channel 1 to channel 10, leveraging the ESP-IDF v5.1 library [1], which aligns with the capabilities of the selected hardware. As the FTM protocol is association-less, we have devised an association-less channel-switching mechanism using the ESPNOW protocol provided by Espressif. This approach is similar to that demonstrated in [32].

To ensure accurate measurements, we have configured the ESP32 CPU frequency to 240 MHz and set the tick rate for FreeRTOS to 1000 Hz. All collected FTM data is transmitted via a serial connection and stored locally on a laptop equipped with Ubuntu 20.04. This setup enables thorough data analysis and further processing to evaluate the system's performance in various indoor scenarios.

## 6 EVALUATION

In this section, we present the comprehensive experimentation conducted to assess the performance of *FTM-Sense*, addressing the following key questions:

**(1)** What is the overall system performance in real-world scenarios? **(2)** How does *FTM-Sense* perform compared to the widely used PIR sensors in detecting static occupants? **(3)** Can *FTM-Sense* deliver accurate results in entirely unfamiliar environments without the necessity for training in the new setting? **(4)** How well does *FTM-Sense* perform in non-line-of-sight (NLOS) scenarios? **(5)** What is the adaptation speed of *FTM-Sense* to new indoor environments?

### 6.1 Methodology

To address the above research questions and evaluate the performance of *FTM-Sense* in diverse real-world scenarios, we construct four baseline models leveraging random forest (RForest), k-nearest neighbors (KNN), XGBoost, and logistic regression (LR) classification algorithms. These models are developed based on occupancy data that we collected over a period of 3 hours in a meeting room in an office building, as illustrated in Figure 3 with dimensions of 255 cm by 410 cm. To ensure consistency and minimize the impact of setup variations, we exclusively used this baseline dataset for training the baseline models and did not use it for testing during any of the subsequent evaluation processes. In this setup, we place two ESP32S2 boards on two opposite sides of the rooms to ensure we cover as much area as possible in the rooms. In addition, for consistent measurements, we set the device's height to 130 cm from the ground, considering the baseline room furniture height. Through testing, we found that these heights provided more accurate measurements when the baseline room was empty, leading us to utilize this height for all experiments.

To optimize energy consumption while maintaining a reasonable resolution for occupancy detection, we adopted a data collection approach where one sample is collected every 20 seconds, resulting in consecutive channel samples being two seconds apart. To achieve this, we configured the burst period to 200 ms, the number of bursts to 8 bursts, and the number of frames to 32 frames [32].

We develop the baseline models using extracted statistical features of variance, standard deviation, and root mean square, as described in Section 4. We extract these features from all of the datasets that we use in this study. Table 3 presents the overall system performance using four baseline models tested in various real-world scenarios. To assess the system's performance overall, we tested the models using 7 hours of randomly selected datasets. The test datasets include the datasets that are collected in 6 completely new rooms, the baseline room with the 2 new configurations, and 2 NLOS scenarios. We explain these scenarios in detail in the following sections. The purpose of reporting these results is to illustrate *FTM-Sense*'s overall performance without focusing on specific scenarios.

As Table 3 demonstrates, RForest and XGBoost perform best with 97.72% and 96.58% accuracy and 97.37% and 96.02% F1-score, respectively, outperforming other models. Due to space limitations, we focus on reporting the performance of these two models in the remaining evaluation experiments.

**Table 3: Overall system performance using four baseline models that are developed based on 3 hours occupancy data collected in a single meeting room. The reported results are based on the datasets that we collected in different experiments, including NLOS data and data collected in 7 different rooms in various scenarios.**

| | **Accuracy** | **F1-score** | **Precision** | **Recall** |
|---|---|---|---|---|
| **RForest** | 97.72 | 97.37 | 98.29 | 96.46 |
| KNN | 83.48 | 77.62 | 95.14 | 65.55 |
| **XGBoost** | 96.58 | 96.02 | 97.69 | 94.41 |
| LR | 74.86 | 71.09 | 71.43 | 70.76 |

### 6.2 Experiments

After fine-tuning the baseline models, we evaluated the system in the following scenarios:

*6.2.1* ***Static Occupant Detection:*** In this phase, we explore the capability of *FTM-Sense*, using the baseline models, to distinguish static occupants from the room's objects without the need for new model training. Static occupant refers to someone who is not moving in the room but still has small movements. These movements are classified as *very fine motion* (e.g., breathing only while sitting) and *fine motions* (e.g., typing on the laptop or bending and straightening back while sitting) [20]. The current occupancy sensing systems in the buildings struggle to detect either of these two motion types.

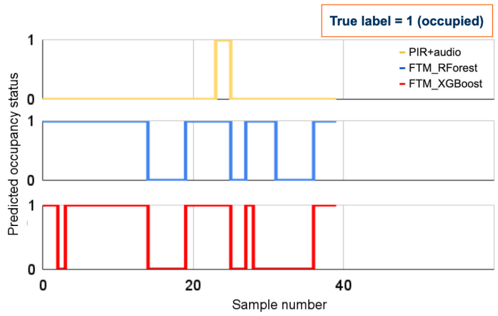
We tested the system's performance in scenarios involving "fine" and "very fine" motions and conducted a comparison with a dual-tech Passive Infrared (PIR) sensor installed in all the office building rooms where we conducted our experiments, as shown in Figure 3.

***Data collection.*** We collected the data for this experiment in the same meeting room where we collected the baseline dataset (Figure 3). During the data collection process, one occupant remained seated in a chair placed in the middle of the table. We collect data in two distinct scenarios to determine the system's static occupancy detection threshold. Firstly, we collect data for 80 minutes (four 20-minute sessions) while the static occupant exhibited very fine motion while watching a movie (no movement other than breathing). In the next 80 minutes (four 20-minute sessions), we collect data while the occupant engages in fine motions, such as typing without significant body movement or bending and straightening back while reading or watching a movie.

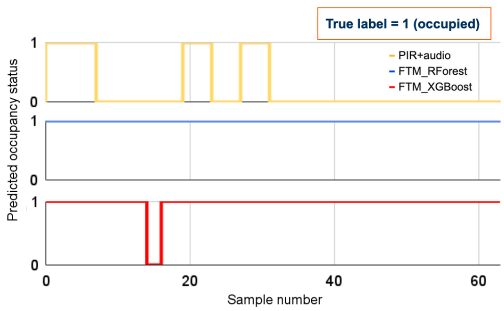
Simultaneously, a dual-tech (PIR+audio) energy-harvesting PIR sensor installed in the room's center ceiling also collected motion data during these periods. We utilized the PIR sensor data for the same timeframes to compare and evaluate the performance of *FTM-Sense* in detecting static occupants.

***Results.*** Our exploratory analysis confirms that static occupants with fine motions have a noticeable impact on the measured RTT values, similar to non-static occupants. Figure 4 illustrates the occupancy detection results of the PIR sensor and *FTM-Sense* during both "fine" and "very fine" motions scenarios.

In the "very fine" motion scenario (Figure 4a), the PIR sensor detects the occupant's presence only once, totaling approximately 90 seconds during the 14 minutes of the occupant's presence in the room. In contrast, our RForest model experienced three detection failures, and the XGBoost model had four failures, totaling less than half of the entire 14 minutes duration. However, in the "fine"




(a) A sample comparison of our FTM-based system with the PIR sensor's performance in the very fine motions scenario.




(b) A sample comparison of our FTM-based system with the PIR sensor's performance in the fine motions scenario.

**Figure 4: Our FTM-based occupancy sensing system outperforms PIR sensor when a static occupant is present in both "fine motions" and "very fine motions" scenarios.**

motions scenario (Figure 4b), the PIR sensor could only detect the occupant's presence three times, accounting for about 6 minutes out of the 20-minute session. On the other hand, our RForest model accurately detected the occupant's presence throughout the entire session (about 20 minutes), with the XGBoost model experiencing only a 20 seconds detection failure.

As depicted in Figure 5, *FTM-Sense* exhibits remarkable performance in scenarios with "fine motions", achieving 97.68% accuracy and 98.83% F1-score using the baseline RForest model. In the case of "very fine motion", both RForest and XGBoost models demonstrated acceptable performance, achieving an accuracy of 70.09% and an F1-score of 83.57%. We should note that the reported results are based on the part of the data that the room was occupied. This approach ensures that the results are not biased since, from our observations, *FTM-Sense* demonstrates very high accuracy in detecting unoccupied rooms. These results overall highlight the effectiveness of *FTM-Sense* in accurately detecting occupancy, particularly in scenarios with static occupants exhibiting fine motions.

***Discussion.*** These experiments depicted that the current system does not have enough resolution to accurately detect the millimeter movement of the human chest due to breathing. However, as the experiment results show, the system is able to detect the occupant while having fine motions like typing. The ability to detect static occupants with fine motions by itself tackles the PIR sensors' weaknesses that are commonly used in most smart buildings.

*6.2.2* ***System Adaptability to New Environments:*** In this part, we investigate *FTM-Sense* robustness in a new environment. First,

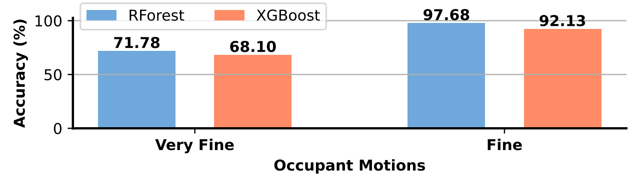



**Figure 5: System performance result in the case of occupant's very fine motions and fine motions.**

we explore the system's adaptability to a change in the baseline room configuration. Second, we test *FTM-Sense* in 3 new rooms in an office building and 3 new rooms in a residential building to find how it performs without further training.

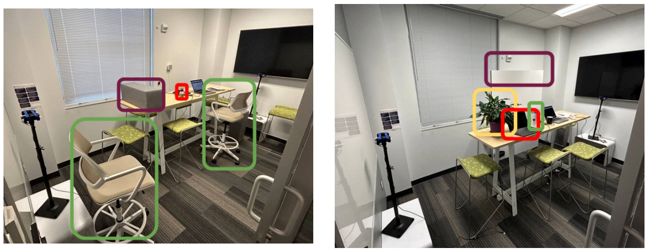
***Data collection.*** This experiment includes two sub-experiments that are discussed below. The location where the participant sits in the rooms is also changed in both sub-experiment. These two sub-experiments are as follows:

(1) **New configurations in the baseline room:** For this part, we collect data in the same room where we collected the baseline dataset. We change the configuration of the room two times and collect 60 minutes of data in each new configuration. Figure 6 shows the two new room configurations where objects in the room and their location is changed.
(2) **Completely new rooms:** We collect 60 minutes of data in three new rooms in the office building and 3 bedrooms in a residential building.

***Results.*** The baseline models are tested on all eight collected datasets to investigate the system's adaptability to a new environment. The results of each of the sub-experiments are discussed as follows:

(1) **New configurations in the baseline room:** Figure 7 demonstrates that the system results in more than 88.89% accuracy and 87.93% F1-score in a new room configurations.
(2) **Completely new rooms:** Figure 8 shows the *FTM-Sense* performance in six new rooms with different sized and configurations. As we can observe, the system performance in these cases is still outstanding, with more than 87.83% accuracy and 89.20% F1-score.

***Discussion.*** *FTM-Sense* demonstrates remarkable adaptability to environmental changes. As Figure 6 depicts, we examined changes in the baseline room configuration by changing the number of electronic devices in the room, adding new objects with different materials from the original room's objects such as whiteboard, chairs, plant, and food to simulate some common configuration changes in the office spaces. The system accurately detects the room occupancy status without any modification or fine-tuning of the classifier models. Also, the system performs reasonably well in new rooms that the model was not initially trained in. As shown in Figure 8, *FTM-Sense* achieves excellent performance in all new rooms, except for the third room in the office building (OF3), where the furniture height differs from the other rooms. This observation suggests that the system's deployment height may play a crucial role in achieving optimal performance. Further investigation into deployment height and its impact on system performance could provide valuable insights for future improvements.



(a) Configuration 1: the table is moved to the side of the room, and two drafting chairs and a cube stand are added to the original room configuration.

(b) Configuration 2: a plant pot, a whiteboard, an additional laptop, and a metal coffee mug are added to the original room configuration.

**Figure 6: Two new room configurations that the system adaptability is tested in them.**

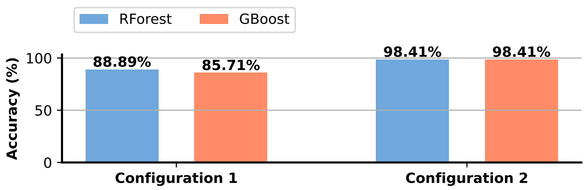



**Figure 7: System adaptability performance which is tested in the original room but with two new configurations that are shown in Figure 6.**

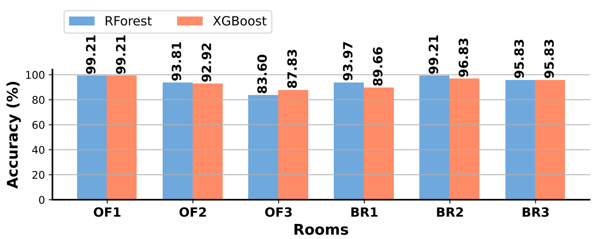



**Figure 8: System adaptability performance which is tested in six completely new rooms.**

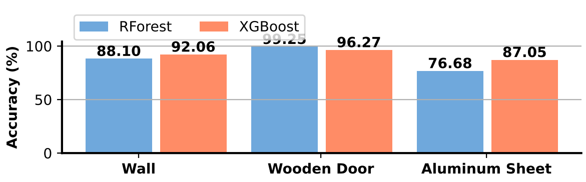


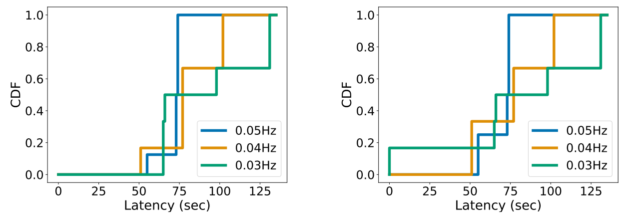

**Figure 9: System performance tested in three different NLOS scenarios.**

### 6.2.3 *FTM-Sense Performance in NLOS Conditions.*

In this section, we assess *FTM-Sense*'s performance under non-line-of-sight (NLOS) conditions, where an object obstructs the direct line of sight between the initiator and the responder. To evaluate the system's robustness in NLOS scenarios, we conducted experiments in three different settings.

***Data collection.*** For this part, we collect data with a room wall, a wooden door, and a wooden door with an aluminum sheet cover

**Figure 10: Wooden door with an aluminum sheet cover used to obstruct the LOS.**




(a) RForest model detection latency

(b) XGBoost model detection latency

**Figure 11: RForest and XGBoost models' change detection latency distribution for 3 different sampling rates.**

(as shown in Figure 10) blocking the LOS. Each scenario involves a data collection duration of 40 minutes.

***Result.*** Figure 9 shows the *FTM-Sense* performance in 3 different NLOS scenarios. As we can observe, the system performance in these cases is still outstanding in the case of wooden door and wall blockage, with more than 92.06% accuracy and 93.15% F1-score. However, in the case of aluminum sheet blockage, the system performance is still acceptable, with more than 87.05% accuracy and 82.76% F1-score.

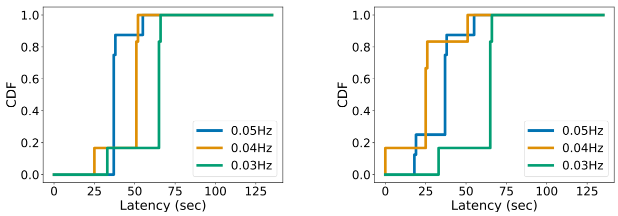
***Discussion.*** *FTM-Sense* demonstrates promising performance in NLOS scenarios, making it a great candidate for various indoor spaces. However, this capability can be a double-edged sword when the system is deployed near a room's wall. Therefore, careful consideration of the initiator and receiver's placement and orientation in the room is crucial to prevent any confusion when occupants pass by the wall from the outside.

### 6.2.4 *FTM-Sense Adaptation Latency.*

In this part, we evaluate the system's occupancy detection latency after a change in the room configuration to understand how quickly *FTM-Sense* can adapt to new settings.

***Data collection.*** To measure the system's adaptation latency, we conduct experiments in the same meeting room where we collected the baseline dataset. During each data session, the room is unoccupied, and a drafting chair is remotely moved from outside the room every 5 minutes for less than 1 minute. We record the timestamp of the chair's movement start and end time. The data is collected for 40 minutes at three different sampling rates: 0.05 Hz, 0.04 Hz, and 0.03 Hz to find the distribution of the system latency in each sampling rate and compare them.

***Results.*** By moving the chair multiple times and estimating the system latency in each experiment, we derive the distribution of *FTM-Sense*'s adaptation latency. Figure 11 presents the cumulative distribution function (CDF) curves of the RForest and XGBoost baseline models' latency. As Figure 11a and 11b show, both models exhibit the same maximum latency, which depends on the sampling rate. The blue lines in both plots represent the current system




(a) RForest model detection latency (b) XGBoost model detection latency

**Figure 12: RForest and XGBoost models' change detection latency distribution for three different sampling rates while the features extracted from 3 consecutive samples instead of 5.**

latency, which is 74 s for maximum latency and 55 s for minimum latency, considering the sampling rate of 0.05 Hz.

***Discussion.*** This experiment shows that *FTM-Sense* can adapt itself to the new room configuration in less than 80 seconds without the need for any extra training effort. However, it is essential to consider the sampling rate and the number of consecutive samples used for feature extraction, as they can influence the system's latency and accuracy. Figure 12 clearly shows that extracting features from fewer consecutive samples (3 samples here) reduces the system latency but the system accuracy might be affected in this case.

Additionally, the adaptation latency can be influenced by the time of the change during the scanning cycle. Changes occurring at the beginning of the scan cycle (e.g., first channel) can be detected faster than changes occurring towards the end (e.g., last channel).

## 7 CONCLUSION

In this study, we explored the potential of Fine Time Measurement (FTM) as a primary sensing technology for achieving accurate and adaptable sensor-free occupancy detection within diverse indoor environments. Our comprehensive testing across diverse indoor environments, including static occupants' presence, line-of-sight (LOS), and non-line-of-sight (NLOS) conditions, affirmed the superiority of FTM over traditional methods.

The results demonstrate that leveraging FTM signal variations, which happens mainly due to multipath reflections caused by human presence, significantly enhances occupancy detection accuracy. *FTM-Sense* shows great potential to revolutionize occupancy sensing technology and significantly impact the future of smart building environments. Further advancements and real-world implementation will unlock even more possibilities for smart buildings, promoting sustainability and optimizing occupant experiences. Moreover, this study can serve as a source of inspiration for researchers to explore the potential of FTM in other human sensing applications.

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